Figure 9.24. The starting positions for the two teams of agents in the MAgent environment. The light squares are the individual agents.

Figure 9.25. This depicts the action space for agents in the MAgent library. Each agent can move in 13 different directions or attack in 8 directions immediately around it. The turn actions are disabled by default, so the action space is 13 + 8 = 21.

Figure 9.26. The structure of the observation tensor. It is an N x 13 x 13 x 7 tensor where N is the number of agents on the team.

Figure 9.27. Screenshot of the MAgent battle game after training with mean field Q-learning. The dark team has forced the light team into the corner and it’s attacking them.

Chapter 10. Interpretable reinforcement learning: Attention and relational models

Figure 10.1. A simple graph. Graphs are composed of nodes (the numberlabeled circles) and edges (the lines) between nodes that represent relationships between nodes. Some data is naturally represented with this kind of graph structure, and traditional neural network architectures are unable to process this kind of data. Graph neural networks (GNNs), on the other hand, can directly operate on graph-structured data.

Figure 10.2. A graph neural network can directly operate on a graph, compute over the nodes and edges, and return an updated graph. In this example, the graph neural network decides to remove the edge connecting the bottom two nodes. This is an abstract example, but the nodes could represent real world variables and the arrows represent causal direction, so the algorithm would be learning to infer causal pathways between variables.

Figure 10.3. Invariance: Rotational invariance is a property of the function such that a rotation transformation of the input does not change the output of the function. Equivariance: Translational equivariance for a function is when applying the translation to the input results in the same output as when you apply the translation after the function has already been performed on the unaltered input.